# Ultrafast Transient Grating Spectroscopy of GaAs

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**Abstract**

Transient-grating spectroscopy separates carrier recombination from spatial diffusion by measuring the decay of a spatially periodic carrier population. A Gaussian fit to the supplied spectrum gives a peak near 788.66 nm and an FWHM bandwidth of 10.92 nm, corresponding to a transform-limited duration of about 84 fs. Fits to the raw autocorrelation traces, using the tutorial's fundamental and second-harmonic fringe terms, give $(81.5 \pm 0.3)$ fs before and $(79.0 \pm 0.1)$ fs after the compressor (repeat-to-repeat standard deviations); the incomplete `vor2` scan is excluded. The local fringe calibration assumes nearly constant delay-scan speed across the fitted peak. The supplied data do not provide enough calibration information to determine hot-carrier cooling time or absolute carrier diffusivity and mobility.

## 1 Theoretical background

An ultrashort pump pulse with photon energy above the band gap creates electron–hole pairs in GaAs. Carrier–carrier scattering first redistributes energy within the carrier population (thermalization); subsequent energy transfer to the lattice is carrier cooling, while recombination removes electron–hole pairs. A transient-grating (TG) measurement adds spatial information: two coherent pump beams interfere to produce a sinusoidal excitation pattern with period $\Lambda$ and wave number $q = 2\pi/\Lambda$. Recombination reduces the total carrier population, while diffusion additionally smooths the spatial modulation. The one-dimensional diffusion–recombination equation in the tutorial is

$$\frac{\partial n(x,t)}{\partial t} = D_a \frac{\partial^2 n(x,t)}{\partial x^2} - \frac{n(x,t)}{\tau_{\mathrm{rec}}}. \tag{1}$$

For a Fourier component $n_q(t)e^{iqx}$ this becomes $\dot{n}_q = -(D_a q^2 + 1/\tau_{\mathrm{rec}})n_q$. Thus, for a single exponential population, the uniform transient-reflectivity component ($q = 0$) and grating component are The two-beam interference and resulting spatial carrier modulation are illustrated in Fig. **??**.

$$S_{\mathrm{pop}}(t) = A_{\mathrm{pop}} e^{-t/\tau_{\mathrm{rec}}} + C_{\mathrm{pop}}, \tag{2}$$

$$S_{\mathrm{TG}}(t) = A_{\mathrm{TG}} e^{-t/\tau_{\mathrm{TG}}} + C_{\mathrm{TG}}, \tag{3}$$

$$\frac{1}{\tau_{\mathrm{TG}}} = \frac{1}{\tau_{\mathrm{rec}}} + D_a q^2. \tag{4}$$

Thus a plot of decay rate against $q^2$ has intercept $1/\tau_{\mathrm{rec}}$ and slope $D_a$. In a photoexcited electron–hole plasma the ambipolar diffusivity, under the non-degenerate Einstein approximation, is $D_a = 2\mu_e\mu_h k_B T/[e(\mu_e+\mu_h)]$ and the corresponding ambipolar mobility is $\mu_a = 2\mu_e\mu_h/(\mu_e+\mu_h)$, so that $D_a = \mu_a k_B T/e$. This mobility is limited by the less mobile carrier. These relations presume a common carrier temperature and an appropriate non-degenerate regime; at high carrier density, many-body and degeneracy effects can change the diffusion coefficient [**?**].

The optical pulse duration can be measured with a second-order autocorrelator based on second-harmonic generation (SHG). The measured intensity is a function of the relative delay

between two pulse replicas. For a Gaussian pulse, the intensity autocorrelation FWHM is $\sqrt{2}$ times the pulse-intensity FWHM. The transform-limited Gaussian time-bandwidth product is $\Delta t \Delta \nu \simeq 0.441$; a larger product can indicate spectral phase/chirp, or measurement and calibration errors. A spectrometer measures wavelength, with $\Delta \nu \simeq c \Delta \lambda / \lambda_0^2$ for a narrow spectrum. This conversion and the time-bandwidth comparison assume that the fitted spectral profile is representative of the pulse spectrum and approximately Gaussian.

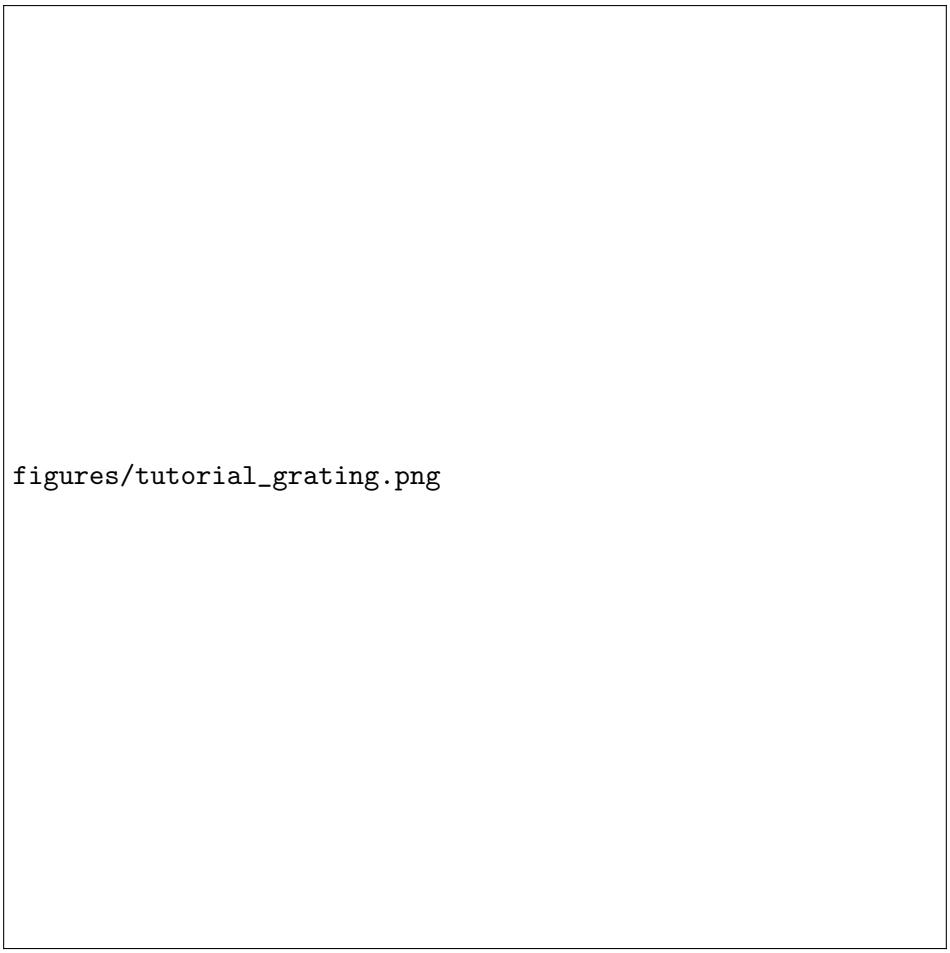



Figure 1: Two crossed pump beams create a sinusoidal excitation pattern in the sample. The grating period $\Lambda_{\mathrm{TG}}$ sets the spatial wave number $q = 2\pi/\Lambda_{\mathrm{TG}}$ used in the diffusion analysis. Reproduced from Fig. 4.5 of the course tutorial.

## 2 Setup

The tutorial describes a Kerr-lens mode-locked Ti:sapphire femtosecond laser. Its intensity-dependent refractive index produces self-focusing that favors pulsed operation; a prism compressor compensates dispersion and reduces pulse chirp. For autocorrelation, a Michelson interferometer creates two pulse replicas with adjustable delay and a BBO crystal generates a second-harmonic signal proportional to their temporal overlap. For the sample measurement, the beam is split into pump and probe paths; two pump beams cross on the GaAs sample and make the excitation grating. Two probe beams arranged in BOXCAR geometry provide a diffracted signal and a local oscillator. Their interference gives a heterodyne term proportional to the diffracted field, improving sensitivity over direct intensity detection. The tutorial describes phase modulation of the local oscillator with a rotating coverslip and pump modulation with a chopper; a lock-in amplifier extracts the component at the chopper frequency. The overall

optical layout is shown in Fig. **??**; the individual Michelson autocorrelator and BOXCAR detection principle are shown in Figs. **??** and **??**. These are the tutorial's apparatus and procedure; the provided data do not document which alignment settings or operating parameters were used in this particular run. The files contain repeated delay-signal samples, a wavelength spectrum, and auxiliary traces named `vor*` and `nach*`. Incidence angles, calibrated grating period, sample orientation, pump fluence, sample temperature, and delay-stage calibration were not supplied.

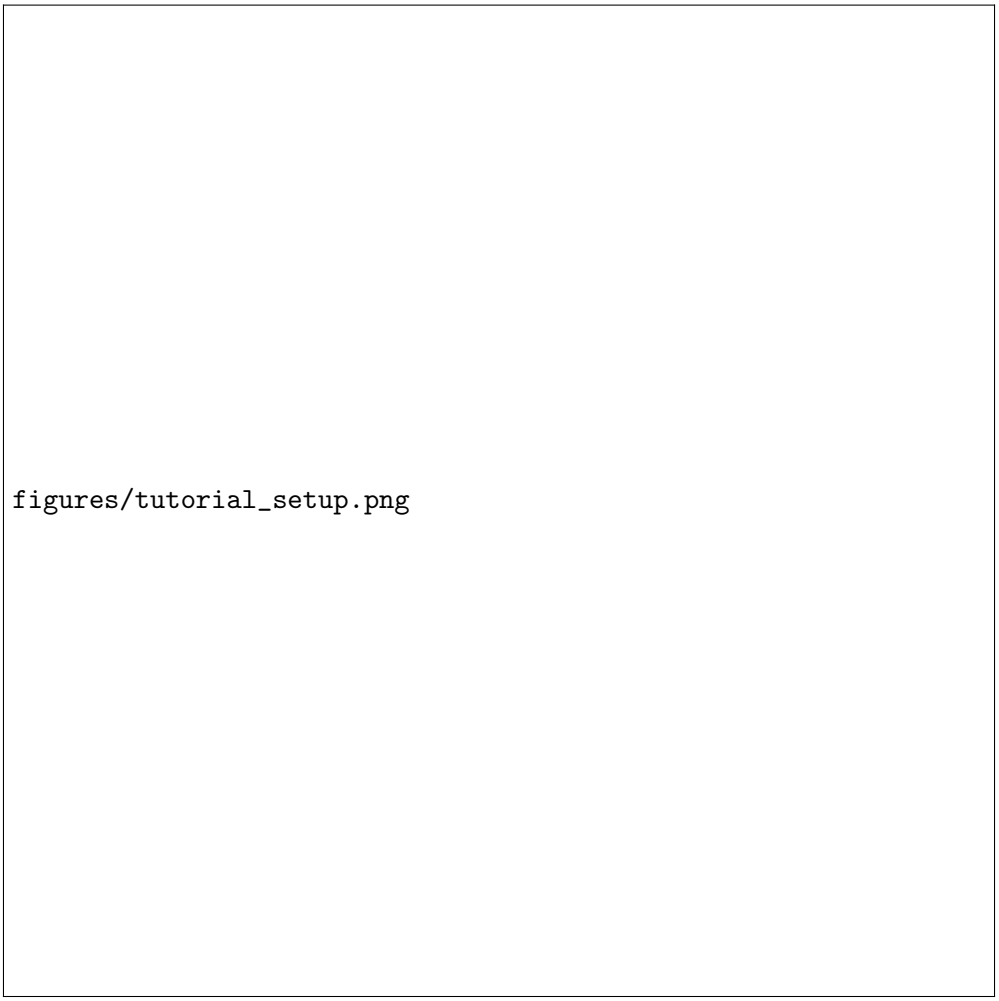



Figure 2: Tutorial schematic of the transient-reflectivity and transient-grating apparatus, including prism compressor, pump and probe paths, delay line, chopper, BOXCAR sample geometry, detector, and lock-in amplifier. Reproduced from Fig. 4.7 of the course tutorial.

## 3 Measurement and analysis

The analysis script `analysis/analyze.py` averages repeated signal columns, estimates pointwise standard errors for plotting, fits the spectrum to a Gaussian plus constant background, and analyzes the autocorrelation traces. TG traces are fitted to a single exponential plus offset over the nominal interval 0.5–15 dataset units. Repeated columns are treated as repeat measurements. The suffixes are provisionally interpreted as grating periods in micrometres only to make the exploratory $q^2$ plot; this interpretation is unverified. The TG delay unit is also unknown, so the fitted lifetimes and rate slope are reported only in dataset coordinates. The autocorrelation files are oscilloscope traces rather than calibrated optical-delay scans; their conversion uses the local optical fringe period and assumes the scan speed is approximately constant over the fitted

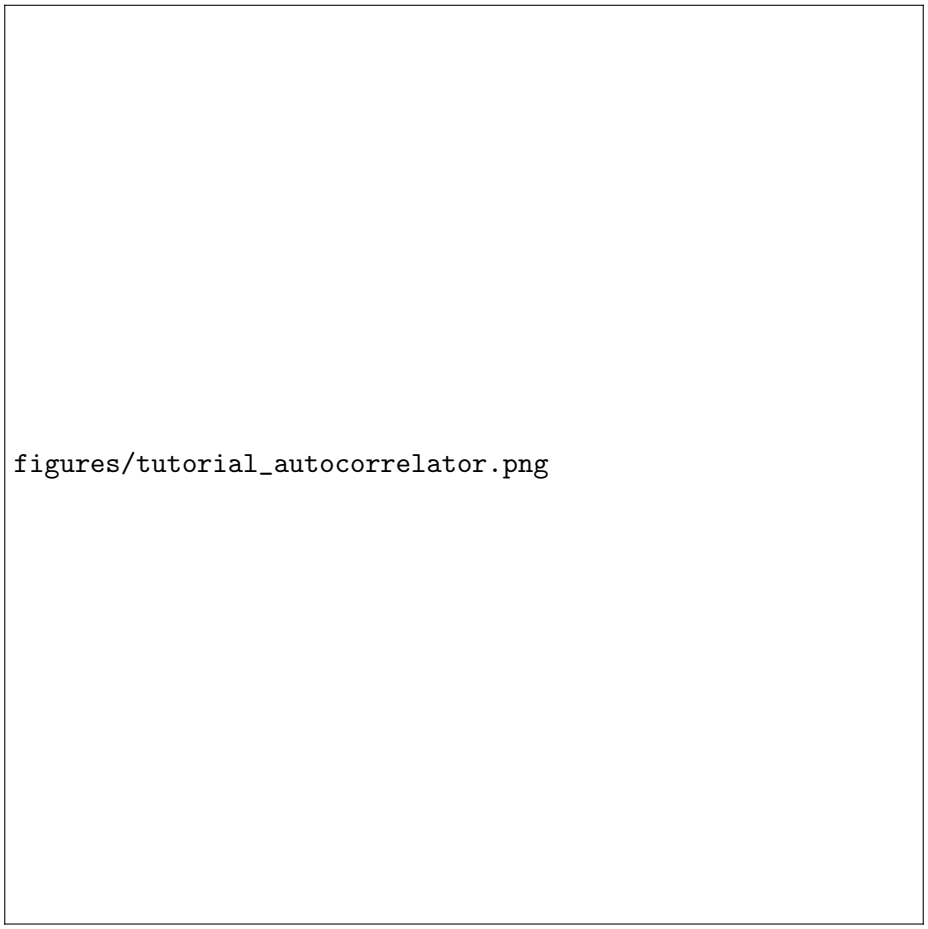



Figure 3: Michelson autocorrelator schematic: the vibrating mirror varies the relative delay between pulse replicas, and a BBO crystal generates the detected second harmonic. Reproduced from Fig. 4.1 of the course tutorial.

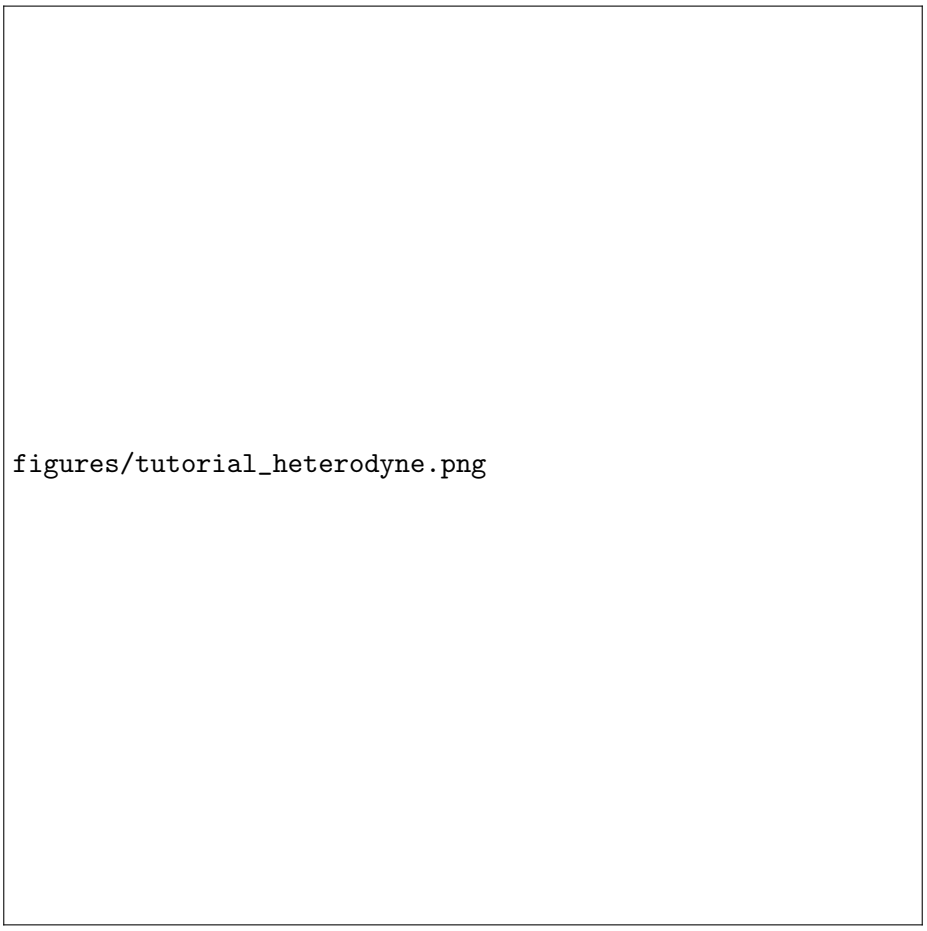



Figure 4: BOXCAR heterodyne detection of the diffracted probe and reflected local oscillator, with the measured signal illustrated as a function of heterodyne phase. Reproduced from Fig. 4.6 of the course tutorial.

peak. The measured second-order autocorrelation is given in the tutorial by Eq. (4.2). With $I(t) = |E(t)|^2$, its full expression is

$$\begin{aligned} I_{\mathrm{AC}}(\tau) = & \int_{-\infty}^{\infty} \left[I^2(t) + I^2(t-\tau)\right] \mathrm{d}t \\ & + \int_{-\infty}^{\infty} [I(t) + I(t-\tau)]\, \mathrm{Re}\{E(t)E^*(t-\tau)\} \, \mathrm{d}t \\ & + \int_{-\infty}^{\infty} \mathrm{Re}\left\{E^2(t)E^{*2}(t-\tau)\right\} \, \mathrm{d}t \\ & + \int_{-\infty}^{\infty} I(t)I(t-\tau) \, \mathrm{d}t. \end{aligned} \tag{4.2}$$

The tutorial identifies these as the constant second-harmonic contribution, the fundamental-frequency interference contribution, the second-harmonic interference contribution, and the intensity autocorrelation, respectively. Following Eq. (**??**), we fit the raw trace to a Gaussian envelope multiplying a constant term and oscillatory terms at $\omega$ and $2\omega$:

$$S(x) = S_0 + e^{-(x-x_0)^2/(2\sigma_x^2)} \left[A_0 + A_1 \cos(2\pi f(x-x_0) + \phi_1) + A_2 \cos(4\pi f(x-x_0) + \phi_2)\right]. \tag{5}$$

The cosine terms follow by separating the optical carrier from its slowly varying pulse envelope. Write the field as $E(t) = a(t)e^{-i\omega_0 t}$, where $a(t)$ is the complex pulse envelope and $\omega_0 = 2\pi c/\lambda_0$. The first field-overlap term in the tutorial's autocorrelation expression then contains

$$\mathrm{Re}[E(t)E^*(t-\tau)] = \mathrm{Re}\left[a(t)a^*(t-\tau)e^{-i\omega_0\tau}\right], \tag{6}$$

which varies with delay at the carrier frequency $\omega_0$. The second-harmonic overlap similarly gives

$$\mathrm{Re}\left[E^2(t)E^{*2}(t-\tau)\right] = \mathrm{Re}\left[a^2(t)a^{*2}(t-\tau)e^{-i2\omega_0\tau}\right], \tag{7}$$

which varies at $2\omega_0$. Integration over $t$ changes the amplitudes and phases but retains these delay oscillations, yielding the two fitted cosine terms. For a locally linear delay scan, $\tau \simeq (\lambda_0/c)f(x-x_0)$, so $\omega_0\tau = 2\pi f(x-x_0)$ and $2\omega_0\tau = 4\pi f(x-x_0)$. The fitted Gaussian intensity-envelope is

$$I_{\mathrm{env}}(x) = A_{\mathrm{env}} \exp\left[-\frac{(x-x_0)^2}{2\sigma_x^2}\right], \qquad A_{\mathrm{env}} = A_0 + |A_1| + |A_2|. \tag{8}$$

In the tutorial's field-envelope notation, its square root is

$$E_{\mathrm{env}}(x) = \sqrt{I_{\mathrm{env}}(x)} = \sqrt{A_{\mathrm{env}}} \exp\left[-\frac{(x-x_0)^2}{4\sigma_x^2}\right], \qquad E_{\mathrm{env}}^2(x) = I_{\mathrm{env}}(x). \tag{9}$$

The appendix plot overlays the normalized $E_{\mathrm{env}}(x)$ (the square root of the fitted intensity-envelope) on the raw signal. The fitted fundamental fringe frequency $f$ in oscilloscope-time coordinates calibrates local delay using the optical fringe period $\lambda_0/c$. Thus $\Delta t_{\mathrm{AC}} = 2\sqrt{2\ln 2}\,\sigma_x f \lambda_0/c$. For a Gaussian pulse, $\Delta t_{\mathrm{pulse}} = \Delta t_{\mathrm{AC}}/\sqrt{2}$. This conversion assumes nearly constant delay-scan speed over the fitted peak. The fitted traces are shown in Fig. **??**. We exclude `vor2`: its recorded scan ends at 2.8 ms, compared with 14 ms for `vor1` and `vor3`, leaving the outer autocorrelation wings incompletely recorded (Fig. **??**). Its sampling interval also differs by a factor of five, so its scan calibration is not directly comparable. Including it would bias the before-compressor pulse-width estimate.

The measured spectrum and Gaussian fit are shown in Fig. **??**. The spectrum file spans approximately 345 nm to 1041 nm. We fitted all recorded points with a Gaussian peak on a constant offset. The fit gives a centre of 788.66 nm, peak intensity of 13 302 a u , median measured intensity of 2501 a u , and fitted half-maximum level of 7902 a u . Its FWHM is $\Delta\lambda \approx 10.92$ nm.

Using the narrow-band conversion $\Delta\nu \simeq c\Delta\lambda/\lambda_0^2$ gives $\Delta\nu \approx 5.26\,\mathrm{THz}$. The Gaussian time-bandwidth product $K = 0.441$ then gives a transform-limited duration $\Delta t_{\mathrm{TL}} = K/\Delta\nu \approx 83.8\,\mathrm{fs}$. The median is the median of the full recorded spectrum, not a local estimate of the spectral baseline; it is shown as a descriptive reference and is not used as the fitted offset.

The direct fits to the raw interferometric autocorrelation traces give a mean pulse duration of $(81.5 \pm 0.3)\,\mathrm{fs}$ before and $(79.0 \pm 0.1)\,\mathrm{fs}$ after the compressor. The uncertainties are sample standard deviations between the two accepted before-compressor scans and the three after-compressor scans; they quantify repeat scatter only, not total measurement uncertainty. The corresponding time-bandwidth products are about 0.43 and 0.42. They are close to the Gaussian transform limit of 0.441, but the apparent compressor-associated change is only about 2.5 fs; without an independent delay-stage calibration and systematic error budget, its significance cannot be established.

# 4 Results

| Quantity | Result from supplied files |
|---|---|
| Gaussian-fit spectral peak | 788.66 nm |
| Gaussian-fit peak intensity | 13 302 a u |
| Measured median intensity (plotted baseline reference) | 2501 a u |
| Fitted half-maximum level | 7902 a u |
| Gaussian-fit spectral FWHM | 10.92 nm |
| Approximate frequency bandwidth | 5.26 THz |
| Gaussian transform-limited pulse duration | 83.8 fs |
| Measured duration before compressor | $(81.5 \pm 0.3)$ fs |
| Measured duration after compressor | $(79.0 \pm 0.1)$ fs |
| Time-bandwidth product before / after | 0.43 / 0.42 |
| Cooling time and electron-phonon coupling | Not determinable from available labeled data |
| $\tau_{\mathrm{rec}}$, $D_a$, and $\mu_a$ | Not reliably determinable |

The individual exploratory TG fit lifetimes and the provisional period interpretation are listed in Table **??**. The $q = 0$ pump–probe file is treated as transient reflectivity; the other suffixes are provisionally read as periods in micrometres. These fits are retained to show what the data files contain, but their time axis is not calibrated in seconds.

Table 1: Exploratory single-exponential fits to the TG and pump–probe traces. Periods and wave numbers use the unverified assumption that filename suffixes are periods in micrometres.

| File | Assumed $\Lambda$ (µm) | $q$ ($\mathrm{m}^{-1}$) | $\tau$ (dataset units) |
|---|---|---|---|
| `GaAs_RT_PP_X.dat` | – | 0 | 1.096 |
| `GaAs_RT_5.6_X.dat` | 5.6 | $1.122 \times 10^6$ | 2.554 |
| `GaAs_RT_6.5_X.dat` | 6.5 | $9.666 \times 10^5$ | 2.725 |
| `GaAs_RT_8_X.dat` | 8.0 | $7.854 \times 10^5$ | 2.960 |
| `GaAs_RT_12_X.dat` | 12.0 | $5.236 \times 10^5$ | 3.238 |

# 5 Discussion

The measured durations are close to the 83.8 fs transform limit, suggesting nearly transform-limited pulses within the analysis uncertainty. The autocorrelation fit uses the tutorial's two

oscillatory terms and assumes a shared Gaussian envelope. Its fringe calibration converts oscilloscope time to optical delay under a locally constant scan-speed assumption. We exclude `vor2` because its shorter scan window truncates the envelope wings and its sampling interval differs from the other before-compressor scans; the file is still displayed for reference in Fig. **??**. Systematic effects from scan nonlinearity, model simplification, and instrument response are not included in the repeat-to-repeat standard deviations. The tutorial distinguishes carrier thermalization (roughly 100 fs) from subsequent energy cooling on a picosecond scale. A time-resolved Raman study of GaAs reported an average electron–LO-phonon scattering time of about 165 fs below an injected density of $10^{17}\,\mathrm{cm}^{-3}$; this microscopic scattering time is not interchangeable with the measured carrier cooling time, which depends on carrier distribution, density, and energy transfer to the lattice [**?**, **?**]. The supplied traces do not identify a calibrated hot-carrier temperature or energy-relaxation observable, so neither a cooling constant nor an average electron–phonon coupling coefficient can be extracted.

The TG delay traces and exploratory single-exponential fits are shown in Fig. **??**. Under the provisional period interpretation, the fitted rate versus $q^2$ regression has slope $-2.98 \times 10^{-13}$ in (dataset unit)$^{-1}\,\mathrm{m}^2$, intercept 0.647 (dataset unit)$^{-1}$, and $R^2 = 0.350$. The negative slope is incompatible with the simple diffusion model, which predicts a non-negative $D_a$, and the fitted TG lifetimes in Table **??** are longer than the $q = 0$ pump–probe lifetime. These inconsistencies show that the present fit/calibration does not recover the expected decomposition. Possible causes include an incorrect interpretation of suffixes or files, uncalibrated delay units, model inadequacy, and additional relaxation channels; the files do not let us select among them. Without verified grating periods and delay calibration, assigning a slope to $D_a$ would introduce an unjustified scale factor. Therefore no numerical recombination time in seconds, diffusivity, or mobility is inferred. Literature shows that ambipolar diffusion can depend on carrier density; for example, calculations for GaAs include density-dependent many-body and degeneracy effects [**?**]. Comparison to reported values would require the sample's excitation density, temperature, calibrated grating periods, and physical time units. A repeat with those calibrations and instrument-response-convolved fits would permit a meaningful determination.

The principal uncertainty sources are local scan-speed variation in the autocorrelation conversion; finite fitting windows and the assumed Gaussian envelope; background and detector-response effects; spectral wavelength/intensity calibration; and unverified meanings and units of the TG files. The reported pulse-width scatter does not include these contributions. The data also omit pump fluence, sample temperature, carrier density, and a physical delay calibration, which are needed for quantitative comparisons across samples.

# 6 Conclusion

The spectrum gives an FWHM bandwidth of 10.92 nm (about 5.26 THz) and a Gaussian transform limit of 83.8 fs. Fits to the raw autocorrelation traces, excluding the truncated `vor2` scan, give $(81.5 \pm 0.3)$ fs before and $(79.0 \pm 0.1)$ fs after the compressor under the local constant-scan-speed assumption. The TG data still lack the calibrations needed for reliable cooling, recombination, diffusivity, and mobility measurements. The accompanying Python script performs the analysis and generates the plots.

# References

[1] Institute of Physics II, University of Cologne, *M2.10: Ultrafast Transient Grating Spectroscopy*, practical-course tutorial, version December 2019, especially Eqs. (4.2)–(4.14) and Secs. 5–6.

[2] J. A. Kash, J. C. Tsang, and J. M. Hvam, "Subpicosecond Time-Resolved Raman Spectroscopy of LO Phonons in GaAs," *Physical Review Letters* **54**, 2151 (1985), doi:10.1103/PhysRevLett.54.2151.

[3] R. F. Leheny, J. Shah, R. L. Fork, C. V. Shank, and A. Migus, "Dynamics of hot carrier cooling in photo-excited GaAs," *Solid State Communications* **31**, 809–813 (1979), doi:10.1016/0038-1098(79)90393-4.

[4] J. F. Young and H. M. van Driel, "Ambipolar diffusion of high-density electrons and holes in Ge, Si, and GaAs: Many-body effects," *Physical Review B* **26**, 2147 (1982), doi:10.1103/PhysRevB.26.2147.

# A Plots

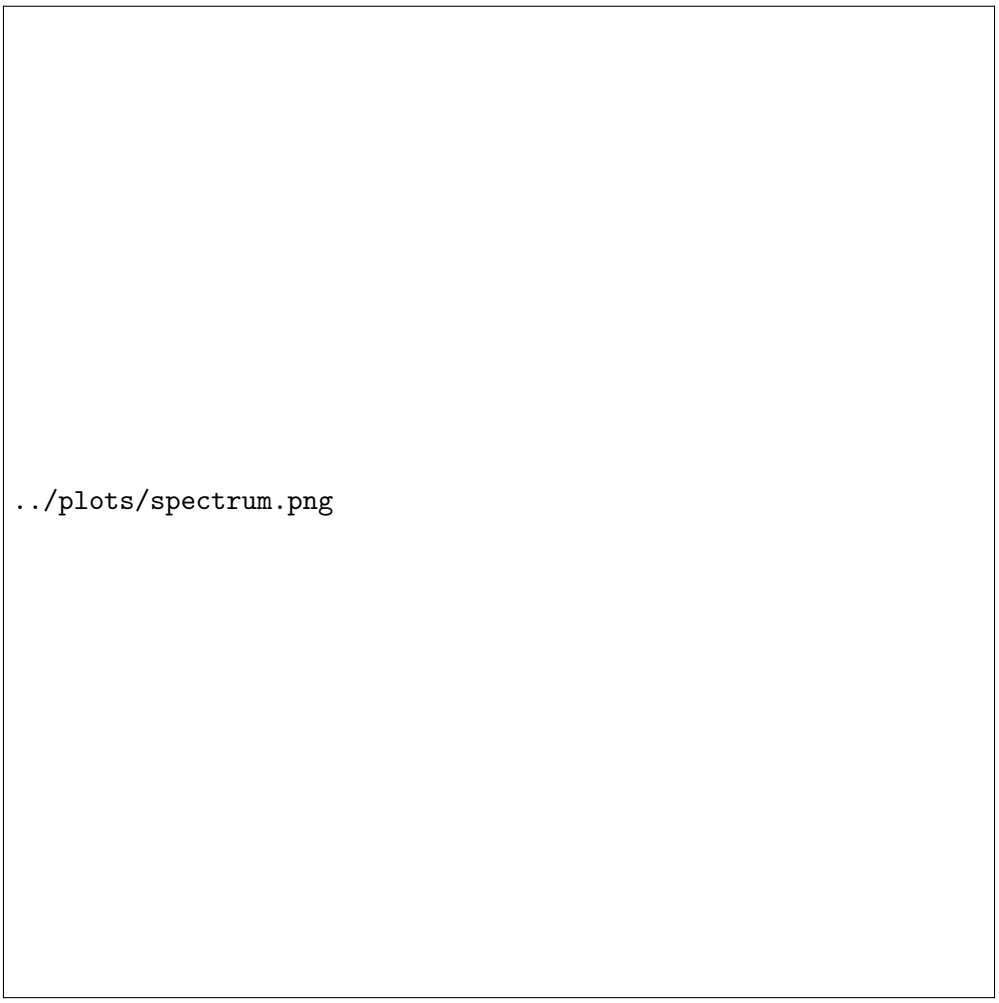



Figure 5: Measured spectrum, Gaussian fit, median level, and fitted half-maximum bandwidth.

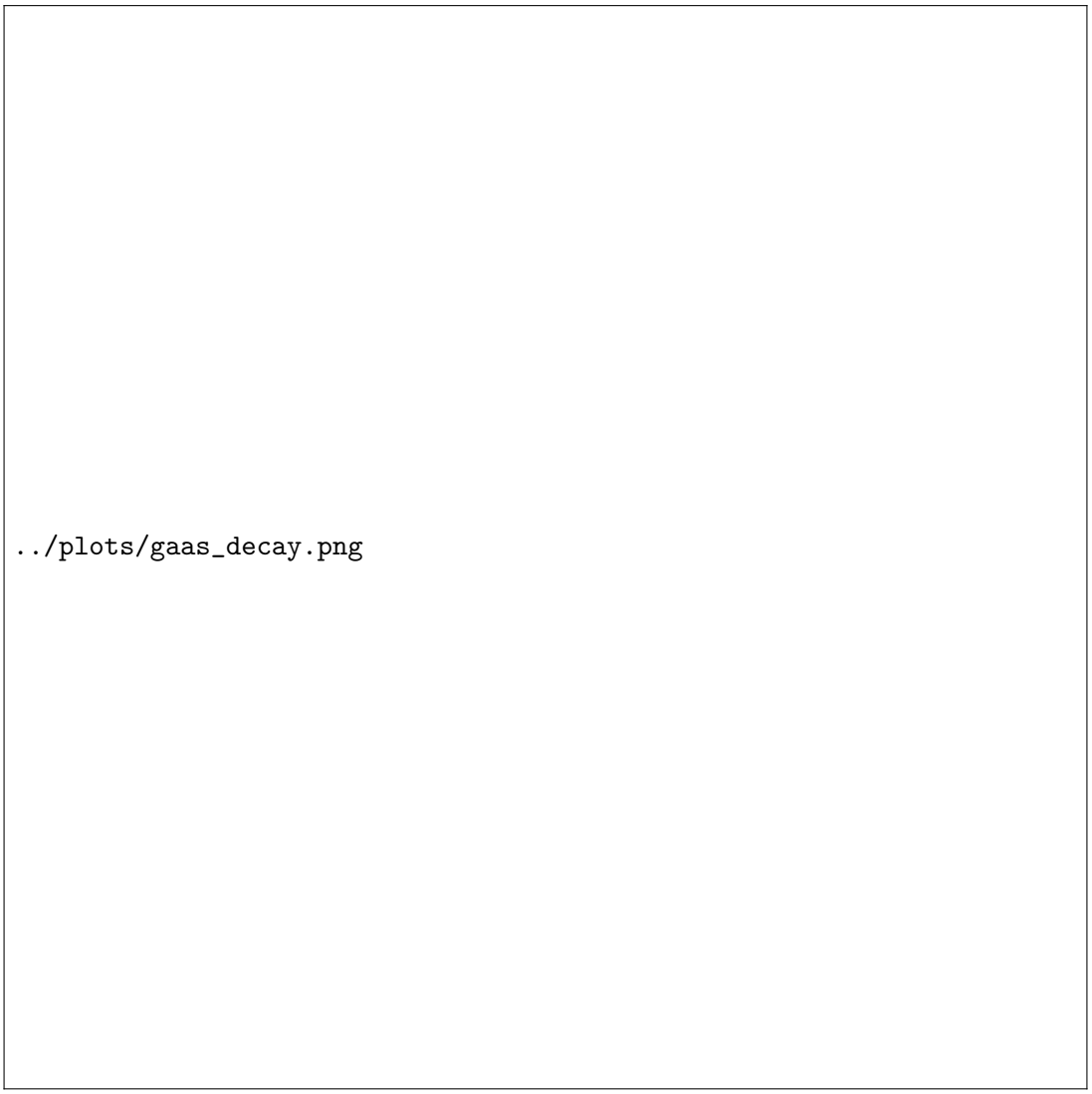



Figure 6: Mean delay traces and exploratory single-exponential fits. Repeated columns are averaged.

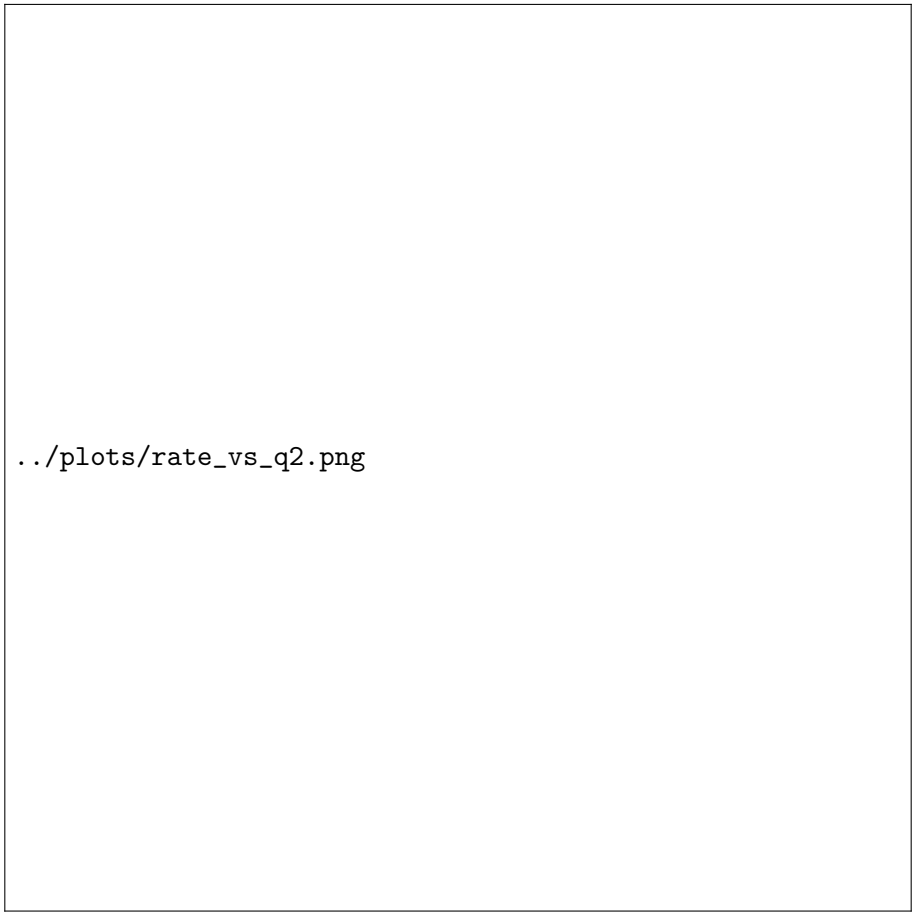



Figure 7: Exploratory fitted rate against nominal $q^2$. The poor linear consistency and missing period calibration prevent extracting a reliable diffusivity.

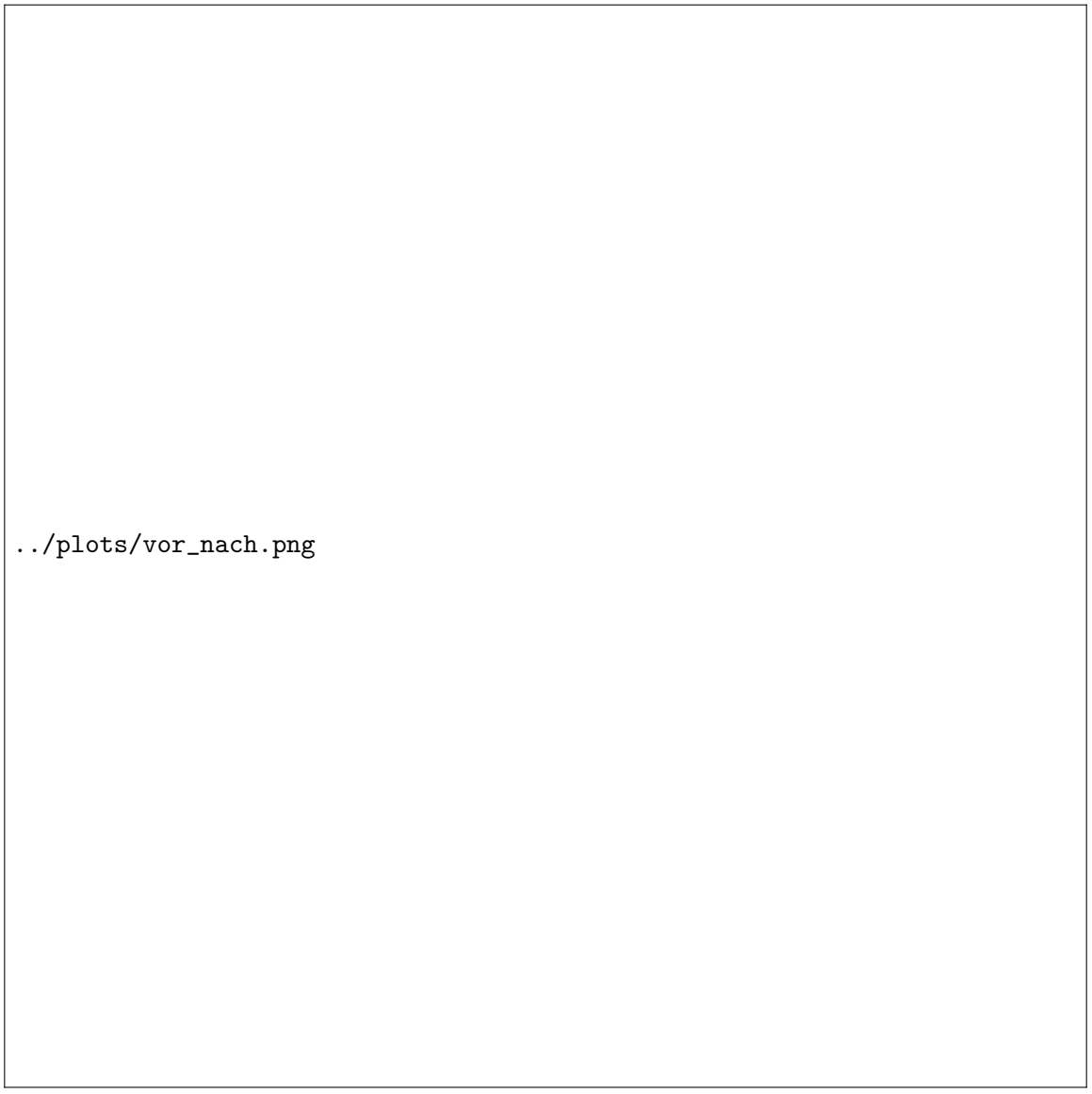



Figure 8: Recorded auxiliary traces grouped as before (vor) and after (nach). The horizontal axis uses recorded time converted to milliseconds.

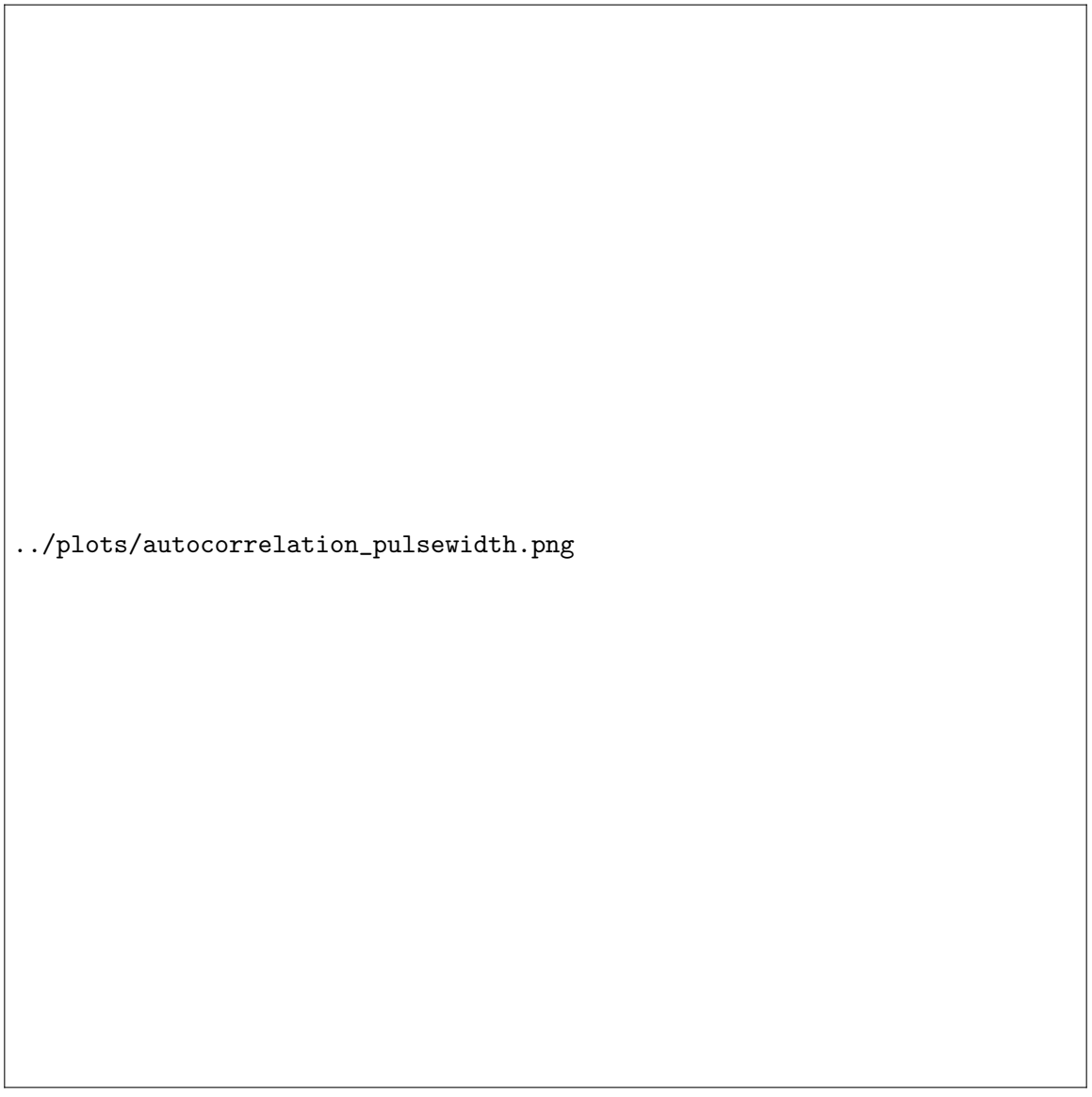



Figure 9: Raw autocorrelation traces after local fringe calibration, with the square root of each fitted Gaussian envelope overlaid. The fit uses the tutorial-based model with $\omega$ and $2\omega$ terms. The truncated `vor2` trace is excluded from these fits.